

Greater Valley Community Services (GVCS)

Executive Summary

Client: Anita Tillman

Student Consulting Team: Isabelle Anis, Terry Kim, Matthew Mingus, Heidi Tao

Background

The James Project, an initiative of Greater Valley Community Services (GVCS), is a community-based safety and training program designed to improve interactions between neurodivergent individuals—particularly those on the autism spectrum—and law enforcement, caregivers, and first responders. Rooted in GVCS's mission to strengthen, educate, and empower families, the program addresses a critical gap in public safety: the lack of practical, experience-based training for real-world, high-stress interactions.

Project Description

Project Opportunity

GVCS currently delivers in-person training and online certification courses focused on autism awareness, communication strategies, and de-escalation. While effective for foundational knowledge, these approaches rely heavily on passive learning and do not adequately prepare participants for unpredictable, real-world scenarios. This creates an opportunity to enhance training through immersive, experiential methods that better simulate high-stress environments.

Project Vision

The vision of this project is to transition the James Project from traditional instruction to an immersive, Virtual Reality (VR)-based training model. This model enables users to engage in realistic, scenario-based learning that builds decision-making skills, empathy, and situational awareness in real time.

Project Outcomes

The project produced a comprehensive VR training framework centered around one high-quality, implementation-ready scenario. This includes:

- **People Outcomes:** Integration of neurodivergent-informed perspectives to ensure authenticity and improve training relevance.
- **Process Outcomes:** Development of a standardized storyboard structure that supports scalable expansion into additional scenarios.
- **Technology Outcomes:** Creation of a VR-based training model that simulates high-stress public interactions, incorporating decision points that influence escalation or de-escalation outcomes.

The scenario specifically focuses on a public environment where sensory overload, communication barriers, and behavioral ambiguity impact interactions between a neurodivergent individual and a responding officer.

Project Deliverables

The primary deliverables include:

- A fully developed VR training storyboard
- Supporting technical documentation for implementation
- A structured decision-tree framework for scenario branching
- Recommendations for future scenario expansion and system scalability

These deliverables are designed to be directly usable by a development team to build the VR experience

Recommendations

This report recommends that GVCS focus on an iterative, development-aligned approach as the VR training experience continues to be built. Close coordination with the PGI development team is essential to ensure that the storyboard and technical implementation remain aligned and realistic.

Additionally, GVCS should establish a structured stakeholder feedback loop, incorporating input from neurodivergent individuals, first responders, and technical experts throughout development to maintain authenticity and accuracy.

The report also recommends stabilizing and validating the current supermarket scenario before expanding into additional scenarios, ensuring that future development builds on a tested framework. At the same time, GVCS should define clear evaluation metrics during development to measure training effectiveness and support scalability.

Finally, GVCS should conduct a controlled pilot of the VR experience prior to broader deployment and leverage AI tools after user interactions to analyze performance data and improve future scenarios.

Student Consulting Team

Isabelle Anis is a senior majoring in Information Systems and Violin Performance. She contributed to user testing, research, documentation, and storyboard development. Isabelle will pursue a Master's in Information Security Policy and Management at CMU after graduation.

Terry Kim is a senior majoring in Information Systems. He contributed to research, solution planning, documentation, and storyboard development. Terry will be building his potato chip startup after graduation.

Matthew Mingus is a junior majoring in Information Systems. He contributed to user testing, storyboard development, client communication, and report writing. He will intern at Capital One over the summer.

Heidi Tao is a senior majoring in Information Systems. She managed the team's progress tracking, meeting prep, and coordination across deliverables. She also contributed to storyboard development and final report work. Heidi is searching for full-time positions after graduation.